Separation and denominator conditions in quadratic prime patterns

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Abstract

We derive a separation property for quadratic prime patterns from the weighted estimates in Sun's simultaneous-avoidance argument. The selected primes can avoid every selected opposite elliptic reduction order while retaining fixed congruences and arbitrarily large prescribed denominator divisibility. We also give a denominator normalization that permits all local weights of order m to be replaced by powers of one weight. These consequences use the cited analytic estimates and elementary valuation arguments; they do not require a further prime-distribution hypothesis.

1 The precise analytic input

Fix distinct integers $c_1, \dots, c_k$, with $k \geq 2$, and put

$$p_j = u + c_j w^2.$$

The fixed congruences, Gaussian ray conditions and reduction-order avoidance conditions in [1, Theorem 4.1] are imposed throughout. A fixed dilation of w can be absorbed into the integers c_j. All such data are chosen before the parameter X tends to infinity.

We use the following quantitative content of [1, Propositions 5.1 and 6.2, Section 6.1]. There are constants $c, C > 0$ and a sufficiently slowly decreasing function $\tau(X) > 0$, with $\tau(X) \to 0$, such that, for

$$T = \lfloor \tau(X)\sqrt{X} \rfloor,$$

the admissible pairs in $X/2 \leq u \leq X$, $1 \leq w \leq T$ carry nonnegative weights $\omega_X(u, w)$ satisfying

$$\sum_{u,w} \omega_X(u, w) \geq cXT, \qquad 0 \leq \omega_X(u, w) \leq C(\log X)^k. \tag{1}$$

The weights vanish outside admissible pairs. The lower bound is after the removal of all self-order and cross-order obstructions, not merely after a finite list of congruences.

For clarity, the order of limits in the cited argument is essential. Its finite cutoff, denoted by L there, is fixed first. The surviving weight is at least

$$\frac{c_1 XT}{8(\log L)^{4k}}$$

for a positive constant c_1 and all sufficiently large X. This gives the first inequality in (1) with a fixed positive constant. Each surviving weight is a product of k prime logarithms, up to fixed normalizations, giving the second inequality. The diagonal construction of τ allows additional lower thresholds which hold eventually for every fixed positive value of τ.

All assertions below that select prime patterns depend on these cited estimates. The elementary statements about weights, integers and valuations are proved here.

2 Removing a thin strip

Lemma 2.1. *Assume (1). Let $0 < \alpha < 1/2$ be fixed. The function τ may be chosen sufficiently slowly that admissible pairs with $w \geq X^{\alpha}$ still have total weight at least $cXT/2$ for every sufficiently large X.*

Proof. The strip $w < X^{\alpha}$ contains at most $X^{1+\alpha} + X$ integer pairs. Its weight is at most

$$C(X^{1+\alpha} + X)(\log X)^k.$$

Choose the diagonal thresholds so that

$$\frac{\tau(X)X^{1/2-\alpha}}{(\log X)^k} \longrightarrow \infty.$$

This is compatible with every prior threshold: for a fixed positive value τ_j, the displayed quotient tends to infinity, so the start of the jth diagonal interval can be increased to meet this additional condition. Also require $\tau(X)\sqrt{X} \geq 2$, so $T \geq \tau(X)\sqrt{X}/2$. The strip weight divided by XT tends to zero. Subtracting it from (1) proves the assertion. □

Theorem 2.2. *For each selected prime p_j, let d_j be a positive integer such that*

$$d_j = p_j + O(\sqrt{p_j}), \qquad 4 \mid d_j,$$

where the implied constant is uniform in j and the pattern. Under the cited analytic input, the pattern can be chosen so that

$$p_i \nmid d_j \qquad (1 \leq i, j \leq k).$$

The fixed congruence and avoidance conditions remain satisfied, all selected primes tend to infinity, and u/w^2 tends to infinity.

Proof. Apply Lemma 2.1 with $1/4 < \alpha < 1/2$. Since $w^2 \leq T^2 = o(X)$, every p_j/u tends to one, uniformly over the selected patterns. The same is true of d_j/u. Consequently $1/2 < d_j/p_i < 2$ for sufficiently large X. Divisibility would therefore imply $d_j = p_i$.

For $i \neq j$, let $c_* := \min_{i \neq j} |c_i - c_j| > 0$. Then

$$|p_i - p_j| \geq c_* w^2 \geq c_* X^{2\alpha},$$

whereas $|d_j - p_j| = O(\sqrt{X})$. Because $2\alpha > 1/2$, equality is impossible for sufficiently large X. For $i = j$, the divisibility $4 \mid d_i$ excludes $d_i = p_i$, since p_i is an odd prime. The other conditions follow from the support of the weights and from $u \geq X/2$, $w^2 = o(X)$. □

For the elliptic curves used in the defining-polynomial constructions, Hasse's bound supplies the displayed estimate for d_j, and the fixed Gaussian congruences make the opposite order divisible by four. Thus Theorem 2.2 gives precisely the separation used to impose additional formal depth on the primary points. The application of this observation to those constructions is already present in [2, Version 9 separation subsection]; the argument above states its quantitative source and permitted limits.

3 Simultaneous denominator preparation

Let S be a finite set of rational primes, and let $a_1, \ldots, a_s \in \mathbb{Q}^{\times}$ be fixed.

Lemma 3.1. *There is a positive integer D such that, whenever u is a prime outside S and w is a positive integer, the numbers*

$$\lambda = \frac{u}{D^2 w^2}, \qquad t_i = a_i \lambda, \qquad \delta_i = t_i - 1$$

satisfy $v_p(\delta_i) = v_p(t_i) < 0$ for every $p \in S$ and every i. The integer D can also satisfy any prescribed finite set of divisibility conditions.

Proof. Choose D with $2v_p(D) > \max_i v_p(a_i)$ for every $p \in S$, and enlarge it to satisfy the other fixed divisibilities. Since $v_p(u) = 0$ and $v_p(w) \geq 0$,

$$v_p(t_i) = v_p(a_i) - 2v_p(D) - 2v_p(w) < 0.$$

The two terms of $t_i - 1$ have distinct valuations, so its valuation equals $v_p(t_i)$. All requirements concern a finite set of primes. □

This choice can be made after fixing the scales and the finite preparation set, but before selecting primes. Multiplying the denominator dilation only changes fixed quadratic coefficients of the contact forms. The coefficient of the zero contact is zero, so u itself is one of the selected primes; the lower cutoff keeps it outside S. This is within the fixed-dilation scope of the cited theorem.

4 Powers of one local weight

For nonzero rational R, δ, define

$$q_m = R^{1-2m}\delta^{-2m-2}, \qquad q = q_1 = R^{-1}\delta^{-4}, \qquad m \geq 1.$$

Proposition 4.1. *The alternative weights $q'_m = q^{2m-1}$ satisfy*

$$q'_m = q_m \delta^{-6(m-1)}.$$

At every valuation where δ is a unit, q'_m and q_m have the same valuation. At every valuation where $v(\delta) \leq 0$, $v(q'_m) \geq v(q_m)$. In particular, a lower bound for $v(q_m D^2)$ remains true with q'_m in place of q_m.

Proof. The exponent of R in q^{2m-1} is $1 - 2m$ and its exponent of δ is $4 - 8m$. Subtracting the exponent $-2m - 2$ in q_m gives $-6(m-1)$. Taking valuations proves the assertions. □

Corollary 4.2. *Consider factors*

$$1 + \sum_{j=0}^{r-1} q_{m_j}^{2^j} D_j^{2^{j+1}}$$

in a local test. At contact places where $v(R)$ is odd and δ is a unit, replacing q_{m_j} by q^{2m_j-1} preserves every valuation and the distinct-valuation argument for its summands. At the finite places prepared by Lemma 3.1, any prescribed positive lower bounds on the summand valuations are preserved. If, as $\lambda \to +\infty$, one has $R \asymp \lambda^{-1}$, $\delta \asymp \lambda$, and all D_j bounded at the real embeddings, the modified factors tend to one.

Proof. The finite-place conclusions follow term by term from Proposition 4.1. At a contact place, the valuation of the jth nonconstant summand is 2^j times an odd integer; these valuations and zero are pairwise distinct. At infinity, $q = O(\lambda^{-3})$, hence $q^{2m_j-1} = O(\lambda^{-3(2m_j-1)})$. Each nonconstant summand therefore tends to zero. □

The corollary is a statement about the specified local tests and their simultaneous preparation. A complete defining-polynomial construction must still prove the pole realization and index-height steps, specify its equations, and check their possible zero strata. Those separate arithmetic requirements are unaffected by this normalization.

References

[1] Z.-W. Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2 (2026), Theorem 4.1, Propositions 5.1 and 6.2, Sections 6.1–8.3. https://arxiv.org/html/2610.20691v2.

[2] $\mathbb{Z}$ *is Diophantine in* $\mathbb{Q}$, Version 13, 7 November 2026, retained Version 9 subsection on separation and asymmetric formal depth. Commit 5bbd74f6a479776a332cbe036809d5dddae63720. https://github.com/elliotglazer/diophantine-z-q.

[3] J. H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009, Chapter V, Theorem 1.1.